

GPIO DRIVER MANUAL

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GPIO_DefinePurpose(GRP, Px, IO)

Sets the direction (Input/Output) for a single specific pin.

```
GPIO_DefinePurpose('A', 3, OUT);
```

BEFORE (DDRA)

0000 0000

AFTER (DDRA)

0000 1000

GPIO_DefinePurpose_Mask(GRP, PxS, PxF, IO)

Sets the direction for a range of pins simultaneously using a mask.

```
GPIO_DefinePurpose_Mask('B', 0, 2, OUT);
```

BEFORE (DDRB)

0000 0000

AFTER (DDRB)

0000 0111

GPIO_WriteDig(GRP, Px, HL)

Writes a logic HIGH or LOW to a single output pin.

```
GPIO_WriteDig('C', 7, HIGH);
```

BEFORE (PORTC)

0000 0000

AFTER (PORTC)

1000 0000

GPIO_WriteDig_Mask(GRP, PxS, PxF, HL)

Sets a specific range of pins to either HIGH or LOW at once.

```
GPIO_WriteDig_Mask('D', 4, 6, HIGH);
```

BEFORE (PORTD)

0 0 0 0 0 0 0 0

AFTER (PORTD)

0 1 1 1 0 0 0 0

GPIO_ToggleDig(GRP, Px)

Inverts the current state of a single pin (HIGH becomes LOW, and vice versa).

```
GPIO_ToggleDig('A', 0);
```

CURRENT (PORTA)

0 0 0 0 0 0 0 1

TOGGLED (PORTA)

0 0 0 0 0 0 0 0

GPIO_ToggleDig_Mask(GRP, PxS, PxF)

Inverts the state of a specific range of pins simultaneously.

```
GPIO_ToggleDig_Mask('B', 4, 7);
```

BEFORE (PORTB)

1 0 1 0 0 0 0 0

AFTER (PORTB)

0 1 0 1 0 0 0 0

GPIO_WriteWholeGRP(GRP, WholePORTReg)

Writes a full 8-bit value directly to the entire port register.

```
GPIO_WriteWholeGRP('C', 0x55);
```

BEFORE (PORTC)

0000 0000

AFTER (PORTC)

0101 0101

GPIO_SuperWrite(GRP, PxS, PxF, Val)

Surgically writes a value into a specific bit-range without affecting other pins.

```
GPIO_SuperWrite('B', 2, 5, 0b1010);
```

BEFORE (PORTB)

110000 11

AFTER (PORTB)

111010 11

GPIO_ReadDig(GRP, Px)

Returns the digital level (0 or 1) of a single input pin.

```
ui8 val = GPIO_ReadDig('A', 4);
```

HARDWARE STATE (PINA)

0001 0000

RETURNED (VAL)

1

GPIO_ReadDig_Mask(GRP, PxS, PxF)

Reads a range of pins and returns the value shifted to the LSB position.

HARDWARE STATE (PINC)

001111 00

RETURNED VALUE (VAL)

0000 1111 (0x0F)

GPIO_SetBit / ClrBit / TogBit

Generic functions for raw register manipulation using pointers.

```
GPIO_SetBit(&TCCR1B, 3);
```

DIRECT POINTER ACCESS

Allows modifying any hardware register (Timers, ADC, etc.) using the same logic.